

**Draft - expert review pending**

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

**Draft translation - not yet human-reviewed / مسوده ترجمه**

# Environmental Science and Ecosystems

Before you can manage pollution, waste or climate change, you need a map of what you are managing. This unit draws that map. It begins with what environmental science is and how planet Earth is structured, then follows the thread that runs through the whole course: human activities and the environment shape each other. It closes with the two living foundations, ecosystems and natural resources, and with the part an individual, including a teacher, can play.

## Unit learning outcomes

By the end of this unit you should be able to:

- Define environmental science, state its scope, and explain why it matters for Pakistan.
- Describe the natural environment using the four-sphere picture of planet Earth.
- Explain the nexus between environmental problems and human activities, with examples.
- Classify environmental issues as global, regional or local, and give a current example of each.
- Explain what an ecosystem is, why ecosystems matter, what threatens them, and how they are conserved.
- Distinguish renewable from non-renewable natural resources and describe how each is managed.

- List concrete individual responsibilities for protecting the natural environment.

## Prerequisite knowledge

None beyond HSC-level general study skills. This is the opening unit of the course. No prior biology or geography beyond HSC is assumed, and none may be assumed.

## In this unit

1. What environmental science studies, and the planet it studies
2. People, activities and environmental problems
3. Ecosystems and natural resources
4. Protecting the environment: the individual's part

## How to use this unit

Each topic is one complete learning cycle: a real classroom situation, an explanation, an activity, a check on your understanding, a summary, a self-assessment, a practicum task, a summative task and further reading. Work the topics in order, because each builds on the one before it. Keep a notebook for the activities: several of them ask you to collect real observations from your own surroundings, and those notes become the raw material for this unit's summative tasks and for later units on pollution and waste.

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